

JavaScript

Module-wise Homework & Interview Questions

Theory · Practical · Top 35 Interview Q&A

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Target	Graduation / PG — Job Preparation
Modules	12 Modules + Top 35 Interview Questions
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JavaScript — Module-wise Homework & Interview Questions

Code	Type	How to Attempt
T	Theory	Write in notebook — explain in your own words.
P	Practical	Code in VS Code, run in browser console/page.
I	Interview	Most asked in JS job interviews — must memorise!

Module 1 — Introduction to JavaScript

[T] Theory Questions

T1. What is JavaScript? What role does it play alongside HTML and CSS?

Ans: _____

T2. What is the difference between client-side and server-side JavaScript?

Ans: _____

T3. Explain the three ways to add JavaScript to an HTML page. Which is best practice and why?

Ans: _____

T4. What is the browser console? How do you open it? Write 5 different console methods.

Ans: _____

T5. What is the difference between `alert()`, `prompt()`, and `confirm()`? Write examples of each.

Ans: _____

T6. What is the difference between `document.write()` and `innerHTML`?

Ans: _____

[P] Practical Exercises

P1. Create an HTML page that shows an alert saying 'Welcome to ForkInfosystems!' when it loads.

Ans: _____

P2. Build a page that asks the user their name using prompt() and displays 'Hello, [name]!' in a heading.

Ans: _____

P3. Create a page with a button that uses confirm() to ask 'Are you sure?' and logs the result.

Ans: _____

P4. Write a JS script that logs your name, age, city, and course using console.log().

Ans: _____

Module 2 — Variables, Data Types & Operators

[T] Theory Questions

T1. What is the difference between var, let, and const? Make a comparison table.

Ans: _____

T2. List all 8 JavaScript data types with an example of each.

Ans: _____

T3. What is the difference between == (loose equality) and === (strict equality)? Give 3 examples.

Ans: _____

T4. What is type coercion in JavaScript? Give 3 surprising examples.

Ans: _____

T5. What is the typeof operator? What does typeof null return and why is it a quirk?

Ans: _____

T6. Explain the nullish coalescing operator (??) with an example.

Ans: _____

T7. What is the difference between null and undefined?

Ans: _____

[P] Practical Exercises

P1. Declare variables for a student: name (const), age (let), marks (let), city (const). Log all.

Ans: _____

P2. Create a temperature converter: ask Celsius via prompt, convert to Fahrenheit, display result.

Ans: _____

P3. Write code to check if a number entered by the user is even or odd using the % operator.

Ans: _____

P4. Demonstrate all arithmetic operators (+, -, *, /, %, **) using two numbers.

Ans: _____

P5. Use typeof to check the type of: 42, 'hello', true, null, undefined, [], {}.

Ans: _____

Module 3 — Control Flow

[T] Theory Questions

T1. What is the difference between if-else and switch? When would you use each?

Ans: _____

T2. Explain the ternary operator with syntax and two examples.

Ans: _____

T3. What is the difference between for, while, and do-while loops?

Ans: _____

T4. What do break and continue do inside a loop? Give one example each.

Ans: _____

T5. What is an infinite loop? How do you avoid it?

Ans: _____

T6. Explain for...of and for...in loops with examples showing when to use each.

Ans: _____

[P] Practical Exercises

P1. Write a program that prints numbers 1 to 50. Multiples of 3: 'Fizz'. Multiples of 5: 'Buzz'. Both: 'FizzBuzz'.

Ans: _____

P2. Create a grade calculator: ask marks, use if-else to print A+/A/B/C/F with message.

Ans: _____

P3. Build a simple login system: ask username and password, allow 3 attempts before logout.

Ans: _____

P4. Use a for loop to calculate the factorial of a number entered by the user.

Ans: _____

P5. Build a multiplication table: ask a number from user, print its table from 1 to 10.

Ans: _____

Module 4 — Functions

[T] Theory Questions

T1. What is the difference between a function declaration and a function expression?

Ans: _____

T2. What are arrow functions? How do they differ from regular functions regarding 'this'?

Ans: _____

T3. What is hoisting in JavaScript? Which function types are hoisted?

Ans: _____

T4. What is a closure? Give a practical example with output.

Ans: _____

T5. What are default parameters? What is the rest (...) parameter?

Ans: _____

T6. What is the difference between parameters and arguments?

Ans: _____

T7. What is the spread operator (...) and how does it differ from rest parameters?

Ans: _____

[P] Practical Exercises

P1. Write a function add(a, b) using declaration, expression, and arrow function styles.

Ans: _____

P2. Create a function isPrime(n) that returns true if n is prime, false otherwise.

Ans: _____

P3. Build a makeCounter() function using closures that has increment, decrement, and reset.

Ans: _____

P4. Write a function that takes a name and optional greeting (default='Hello') and returns a message.

Ans: _____

P5. Create a function that uses rest parameters to find the sum of any number of arguments.

Ans: _____

Module 5 — Arrays

[T] Theory Questions

T1. List 10 array methods and explain what each does and what it returns.

Ans: _____

T2. What is the difference between `map()` and `forEach()`?

Ans: _____

T3. What does `filter()` return? How is it different from `find()`?

Ans: _____

T4. Explain `reduce()` with a step-by-step example calculating sum of an array.

Ans: _____

T5. What is array destructuring? Give 3 examples including rest and default values.

Ans: _____

T6. What is the difference between `sort()` default behaviour and numeric sort? Why does this matter?

Ans: _____

[P] Practical Exercises

P1. Given marks = [45,78,23,89,90,56,12,88], find: highest, lowest, average, and all above 60.

Ans: _____

P2. Filter an array of students (objects with name & marks) to get those who passed (≥ 40).

Ans: _____

P3. Use `map()` to create a new array of full names from [{firstName, lastName}] objects.

Ans: _____

P4. Use `reduce()` to group students into 'passed' and 'failed' arrays.

Ans: _____

P5. Sort an array of products by price (low to high). Then reverse it.

Ans: _____

Module 6 — Objects

[T] Theory Questions

T1. What is an object in JavaScript? How do you create one?

Ans: _____

T2. What is the difference between dot notation and bracket notation for accessing properties?

Ans: _____

T3. What does the 'this' keyword refer to inside an object method?

Ans: _____

T4. Explain `Object.keys()`, `Object.values()`, and `Object.entries()` with examples.

Ans: _____

T5. What is object destructuring? How do you rename a variable while destructuring?

Ans: _____

T6. What is optional chaining (`?.`)? Why is it useful?

Ans: _____

T7. What does `Object.freeze()` do? Give a use case.

Ans: _____

[P] Practical Exercises

P1. Create a student object with name, age, marks, and methods: `greet()`, `isPassed()`, `getGrade()`.

Ans: _____

P2. Write a function that takes two objects and merges them using the spread operator.

Ans: _____

P3. Use for...in to loop through an object and display all its key-value pairs.

Ans: _____

P4. Destructure a nested object: {name, address: {city, pin}} and log city and pin.

Ans: _____

P5. Create a product catalog using an array of objects. Add methods to find by name and filter by price range.

Ans: _____

Module 7 — DOM Manipulation

[T] Theory Questions

T1. What is the DOM? How does it relate to HTML?

Ans: _____

T2. What is the difference between getElementById and querySelector?

Ans: _____

T3. What is the difference between innerHTML and textContent? When should you use each?

Ans: _____

T4. Explain classList.add(), classList.remove(), and classList.toggle() with examples.

Ans: _____

T5. What is the difference between setAttribute() and direct property access (e.g. img.src)?

Ans: _____

T6. What methods do you use to create and add new elements to the DOM?

Ans: _____

[P] Practical Exercises

P1. Build a page where clicking a button changes the background colour randomly.

Ans: _____

P2. Create a character counter: as user types in a textarea, show remaining chars (max 200).

Ans: _____

P3. Build a dynamic list: input a name, click Add — it appears in a list. Click list item to remove it.

Ans: _____

P4. Create a page that generates a table from a JavaScript array of student objects.

Ans: _____

P5. Build a simple image gallery: clicking thumbnails shows a larger image in a preview area.

Ans: _____

Module 8 — Events

[T] Theory Questions

T1. What is the difference between `addEventListener` and using `onclick` attribute?

Ans: _____

T2. What is event bubbling? How do you stop it?

Ans: _____

T3. What is event delegation? Why is it better than attaching listeners to each child?

Ans: _____

T4. What does `event.preventDefault()` do? Give a real example.

Ans: _____

T5. What properties does the event object have? List at least 8.

Ans: _____

T6. What is the difference between mouseenter and mouseover events?

Ans: _____

[P] Practical Exercises

P1. Build a click counter: button shows how many times it has been clicked.

Ans: _____

P2. Create a form with name, email, password. Validate on submit: all fields required, email must have @.

Ans: _____

P3. Build a live search: type in input, list filters matching items from an array.

Ans: _____

P4. Create a keyboard event: press 'd' to toggle dark mode, press 'r' to reset page.

Ans: _____

P5. Build a drag-highlight effect: when mouse enters a card, it glows. When it leaves, glow disappears.

Ans: _____

Module 9 — ES6+ Modern JavaScript

[T] Theory Questions

T1. What are template literals? How are they different from regular string concatenation?

Ans: _____

T2. What is a Promise in JavaScript? What are its three states?

Ans: _____

T3. What is async/await? How does it improve on using .then() chains?

Ans: _____

T4. What is the Fetch API? Write a GET request example with error handling.

Ans: _____

T5. What is the difference between Promise.all() and Promise.race()?

Ans: _____

T6. What is an ES6 module? What is the difference between named export and default export?

Ans: _____

[P] Practical Exercises

P1. Write an async function that fetches user data from jsonplaceholder.typicode.com/users and logs names.

Ans: _____

P2. Create a function using fetch to POST data (name, email) to an API and log the response.

Ans: _____

P3. Build a page with a Search button: fetch results from an API and display them in cards.

Ans: _____

P4. Write a Promise that resolves after 3 seconds and logs 'Done!' using both .then() and async/await.

Ans: _____

P5. Create an error-handling wrapper: if fetch fails, show a user-friendly error message on the page.

Ans: _____

Module 10 — Object-Oriented Programming

[T] Theory Questions

T1. What is OOP? Name and explain the four pillars: Encapsulation, Inheritance, Polymorphism, Abstraction.

Ans: _____

T2. What is the difference between a class and an object (instance)?

Ans: _____

T3. What is the constructor() method in a class? When does it run?

Ans: _____

T4. What are static methods? How are they different from instance methods?

Ans: _____

T5. What is inheritance? How do extends and super work in JavaScript?

Ans: _____

T6. What are private fields (#) in JavaScript classes? Why are they used?

Ans: _____

[P] Practical Exercises

P1. Create an Animal class with name, sound, and speak() method. Extend it with Dog and Cat classes.

Ans: _____

P2. Build a BankAccount class: private balance, deposit(), withdraw(), getBalance(). Validate amounts.

Ans: _____

P3. Create a Library class that manages an array of Book objects: add, remove, search by title.

Ans: _____

P4. Implement a Shape class with area() and perimeter(). Extend with Circle and Rectangle.

Ans: _____

P5. Build a Student class hierarchy: Person → Student → GraduateStudent with appropriate properties.

Ans: _____

Module 11 — Error Handling & Storage

[T] Theory Questions

T1. What are the 5 types of JavaScript errors? Give an example of each.

Ans: _____

T2. What is the purpose of try, catch, and finally? Explain with a code example.

Ans: _____

T3. When would you use throw? How do you create a custom error class?

Ans: _____

T4. What is JSON? What are JSON.stringify() and JSON.parse()? What types can't JSON handle?

Ans: _____

T5. What is the difference between localStorage and sessionStorage?

Ans: _____

T6. What happens if you try to store an object directly in localStorage without JSON.stringify()?

Ans: _____

[P] Practical Exercises

P1. Write a divide(a, b) function that throws RangeError if b===0 and TypeError if args aren't numbers.

Ans: _____

P2. Create a form that saves user preferences (name, theme, font-size) to localStorage and loads on refresh.

Ans: _____

P3. Build a shopping cart that persists in localStorage: add items, show total, clear cart.

Ans: _____

P4. Write a function that safely parses JSON with try-catch and returns null on invalid JSON.

Ans: _____

P5. Build a 'Remember Me' login: save username in localStorage if checkbox is checked, load on next visit.

Ans: _____

Top 35 JavaScript Interview Questions

[I] Interview Questions

I1. What is JavaScript? How is it different from Java?

Ans: _____

I2. What is the difference between var, let, and const?

Ans: _____

I3. What is hoisting? Does it apply to let and const?

Ans: _____

I4. What is the difference between == and ===?

Ans: _____

I5. What is a closure? Explain with a practical example.

Ans: _____

I6. What is the event loop? How does JavaScript handle asynchronous code?

Ans: _____

I7. What is a Promise? What are its three states?

Ans: _____

I8. What is the difference between async/await and .then()/.catch()?

Ans: _____

I9. What is the difference between null and undefined?

Ans: _____

I10. What is NaN? How do you check if a value is NaN?

Ans: _____

I11. What is the difference between `forEach()` and `map()`?

Ans: _____

I12. What does the `filter()` method do? Give an example.

Ans: _____

I13. What is the `reduce()` method? Explain with a sum example.

Ans: _____

I14. What is destructuring in JavaScript? Give array and object examples.

Ans: _____

I15. What are arrow functions? How do they differ from regular functions?

Ans: _____

I16. What is the spread operator `(...)` and rest parameter? How are they different?

Ans: _____

I17. What is the DOM? How is it different from HTML?

Ans: _____

I18. What is event delegation? Why is it used?

Ans: _____

I19. What is event bubbling and event capturing?

Ans: _____

I20. What does `event.preventDefault()` do? Give a real use case.

Ans: _____

I21. What is the difference between `innerHTML` and `textContent`?

Ans: _____

I22. What is `localStorage`? How is it different from `sessionStorage` and cookies?

Ans: _____

I23. What is JSON? What are JSON.stringify() and JSON.parse()?

Ans: _____

I24. What is the this keyword? How does its value change in different contexts?

Ans: _____

I25. What is prototypal inheritance?

Ans: _____

I26. What is the difference between a class and a constructor function?

Ans: _____

I27. What are ES6 modules? What is named vs default export?

Ans: _____

I28. What is the Fetch API? Write a basic GET request.

Ans: _____

I29. What is the difference between synchronous and asynchronous JavaScript?

Ans: _____

I30. What is a callback function? What is 'callback hell'?

Ans: _____

I31. What are higher-order functions? Give 3 examples.

Ans: _____

I32. What is memoisation? Write a simple example.

Ans: _____

I33. What is the difference between deep copy and shallow copy of an object?

Ans: _____

I34. What is optional chaining (?.) and nullish coalescing (??)?

Ans: _____

I35. What is the difference between `call()`, `apply()`, and `bind()`?

Ans: _____
